Love of Variety, Habit Formation, and Dynamic Advertising

A Structural Demand Model with Prior-Informed Preferences

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Roadmap

1. Motivation
2. Model
3. Simulation Design
4. Results: Consumer Behavior
5. Next Steps

Why Does Variety Matter?

- Canonical logit models ignore consumer preferences for variety, which are empirically important (e.g., McFadden 1974; Berry, Levinsohn, Pakes 1995)
- Variety-seeking and habit formation are two sides of the same coin: consumers either prefer to try new things or stick to what they know
- These preferences shape consumer switching behavior, which in turn shapes firms' incentives to target advertising and build market share

Research Question

How much do consumer tastes for variety and habit influence consumption dynamics and firms' advertising strategies in a dynamic market?

What This Paper Does

1. **Model:** Derive a discrete-choice demand system where the utility of a product depends on both deviation from the consumer's experienced mean $\bar{X}_{it}$ and product quality X_{jt}
2. **Simulation:** Characterize consumer and firm behavior across preference regimes (β_i, γ_i) and advertising specifications
3. **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
4. **Counterfactuals (coming):** Study targeting policy and welfare implications

Today's talk: Model + simulation results

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Consumer's Problem

Consumer i chooses product j in period t to maximize:

$$U_{ijt} = \underbrace{\beta_i X_{jt}}_{\text{quality gain}} + \underbrace{\gamma_i \Sigma_{ijt}^2}_{\text{variety component}} + \underbrace{\kappa\, a_{jt}}_{\text{advertising}} - \underbrace{\alpha\, p_{jt}}_{\text{price}} + \xi_j + \varepsilon_{ijt}$$

where $\Sigma_{ijt} = X_{jt} - \bar{X}_{it}$ and $\bar{X}_{it} = \frac{1}{t}\sum_{k=1}^{t} X_{ikt}$

Key parameters:

- $\beta_i > 0$: marginal utility of quality
- $\gamma_i > 0$: variety-seeking
- $\kappa > 0$: advertising shifter

Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from $\varepsilon_{ijt} \sim$ Gumbel$(0, 1)$

State Variable: Experienced Mean

Experienced Mean $\bar{X}_{it}$

$$\bar{X}_{it} = \frac{1}{t} \sum_{k=1}^{t} x_{ikt}$$

Updates each period as the consumer makes a new choice. This is the key state variable linking past behavior to current utility.

Two initialization strategies (both simulated today):

1. **Set mean:** $\bar{X}_{i0} = \bar{J}$, the mean of the product set. Simple benchmark.
2. **Initial state:** Simulate $T_{\text{prior}} = 5$ pre-sample periods from $J \in [1, 2, 3, 4, 5]$ to generate an experience-based prior. Similar to Bronnenberg, Dubé, Joo (2021).

Prior mean yields more heterogeneous behavior — more on this shortly.

Firm's Problem

Firm j solves a dynamic program:

$$V_j(t; \Sigma_{jt}, \lambda_t(\gamma_i)) = \max_{a \in \{0,1\}} \Big\{ \pi_{jt}(a) + \beta_{df}\, \mathbb{E}\big[V_j(t{+}1; \cdot)\big] \Big\}$$

Per-period profit:

$$\pi_{jt}(a) = s_{jt}(p - mc) - \mu\, a$$

Two advertising specifications:

- **Fixed:** $a \in \{0, 1\}$ (binary on/off)
- **Heuristic** $a_t \propto (\bar{X}_{it} - X_{jt-1})^2$ — higher spending when product is far from consumer's mean — work in progress: not ideal measure of drift

Solved via backward induction; firm chooses a to maximize discounted profits (to be completed)

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Simulation Parameters

Table: Parameters

Param.	Value	Description
T	100	Periods
J	5	Products
S	1,000	Simulations
κ	0	Advertising (off)
α	0	Price (off)
ξ_j	0	Quality (off)
β_{df}	0	Discount factor (off)

Preference for Characteristic j:

- $\beta_i \in \{1, 2.5, 4\}$ — low, medium, high quality sensitivity

Preference for Variety:

- $\gamma_i \in \{0.5, 1, 1.5\}$ — low, medium, high variety-seeking

9 regimes $\times$ 2 mean specs $\times$ 2 ad specs $=$ 36 cells

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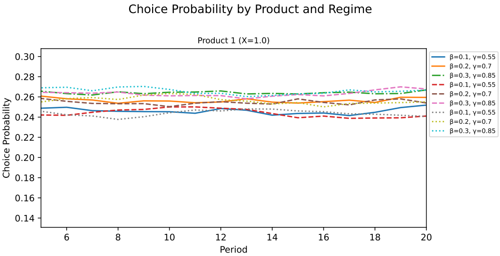
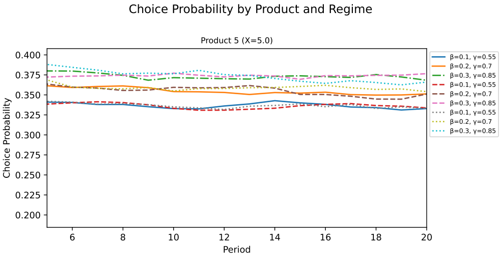
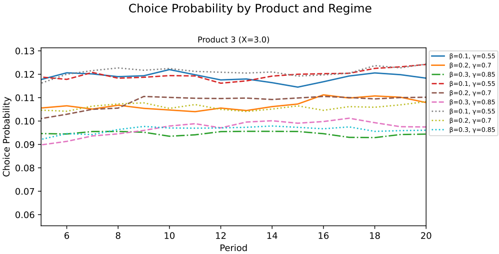
Set Mean Specification







Prior Mean Specification

Product 1:

- Chosen by variety-seeking consumers (γ_i high) who are drawn to the extreme initial state,
- avoided by quality-sensitive consumers (β_i high) who learn that it is low quality.

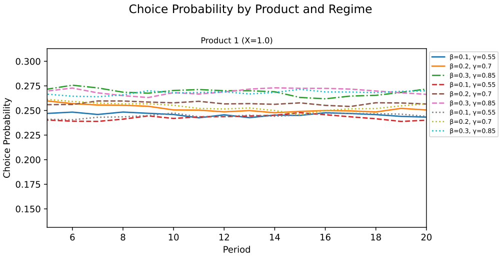
Product 3:

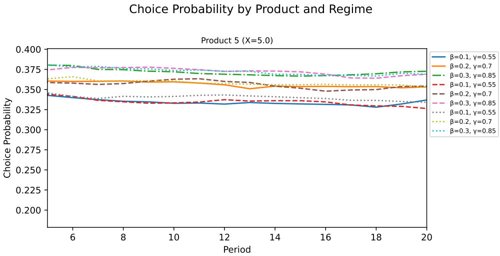
- Chosen by the regime who is neither quality-sensitive nor variety-seeking (low β_i, low γ_i) who are content to stick to the mean.
- Small uptake by medium qualtiy-sensitive consumers (medium β_i, low γ_i), attempt to learn about quality but not variety-seeking enough to try other products.

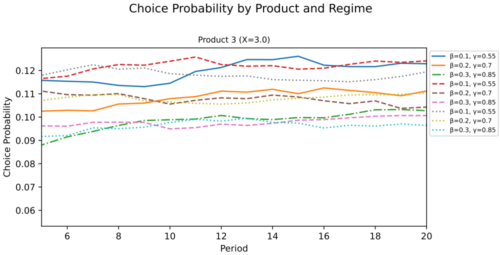
Product 5:

- Chosen by quality-sensitive consumers (β_i high) who are drawn to the extreme initial state and learn that it is high quality.
- Avoided by variety seeking consumers (γ_i high) and those who are neither (β_i low, γ_i low) who are content to stick to the mean.
- Highest probability of choice overall, consistent with idea of utility maximization

Initial State Specification







Initial State Specification

Product 1:

- Chosen by variety-seeking consumers (γ_i high)
- Avoided by quality-sensitive consumers (β_i high), conditional on low variety-seeking

Product 3:

- Chosen by the regime who is neither quality-sensitive nor variety-seeking (low β_i, low γ_i)
- Still avoided by most consumer types

Product 5:

- Almost identical to set mean spec
- Utility maximizing consumers always prefer the highest quality product, only impact probability of choice each period

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Road to Estimation

1. **Multiple consumers**
   Extend simulation to $I = 5$ heterogeneous consumers; aggregate to market shares
2. **Heterogeneous γ_i**
   Draw $\gamma_i \sim N(0, \sigma_\gamma)$; simulate over $\sigma_\gamma \in \{0.5, 1.0, 1.5\}$
3. **Consumer value function**
   Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values
4. **Nielsen scanner data**
   Match simulated moments to data; identify (β_i, γ_i) using BLP instruments
5. **Firm's Problem**
   Ad targeting policy; profit maximization; counterfactuals

Literature Connections

Demand side

- McFadden (1974): MNL foundation
- Berry, Levinsohn, Pakes (1995): RC logit
- Erdem & Keane (1996): Learning dynamics

Advertising / firm side

- Nerlove & Arrow (1962): Goodwill
- Ackerberg (2001): Informative ads
- Goldfarb & Tucker (2011): Targeted digital ads

Contribution

I connect the love-of-variety demand literature to the dynamic targeting literature by making $(X_{jt} - \bar{X}_{it})$ the sufficient statistic for both consumer switching and firm ad decisions, as well as by introducing a novel way to generate consumer heterogeneity in this framework via the love-of-variety.

Summary

1. **Model:** Consumer utility depends on deviation from experienced mean; nests variety-seeking and habit formation
2. **Set mean:** Flat choice probabilities — good benchmark, insufficient for data
3. **Prior mean:** Rich behavioral dynamics; regime differences generate cross-sectional sorting consistent with scanner data
4. **Firm:** Markovian advertising incentive rises when consumer drifts; generates time-varying promotional intensity
5. **Next:** Multiple consumers, structural estimation, Nielsen data

Thank you — questions welcome

Appendix

Simulation Algorithm (Consumer)

1. Draw $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$ for all i, j, t
2. Initialize $\bar{X}_{i0}$ (set mean or prior mean)
3. For $t = 1, \ldots, T$:
   1. Compute $U_{ijt} = \beta_i X_{jt} - \gamma_i (X_{jt} - \bar{X}_{it})^2 + \kappa a_{jt} + \varepsilon_{ijt}$
   2. Set $j^* = \arg\max_j U_{ijt}$
   3. Update $\bar{X}_{it+1} = \frac{1}{t} \sum_{k \le t} x_{ijk^*}$
4. Compute CCPs via log-sum-exp for numerical stability: $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
5. Average over $S = 1{,}000$ simulation draws